

Gauge-Theoretic Signed-Hypergraph Learning

HSiKAN fibers, rotor connections, spike-walk holonomy, and the semiring convolution that unifies them

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2026-06-25 (technical report; conjectures marked as such)

Abstract

We develop a gauge-theoretic account of computation on signed hypergraphs and use it to organise — and extend — the HSiKAN line. Three claims structure the report. (i) Every computation HyMeKo performs over its signed hypergraph is a *semiring convolution* of the signed incidence with a vertex field; the choice of semiring (additive, tropical, parity, fuzzy, spinor) is the choice of computational regime, and we give the conditions under which it is well defined. (ii) The *signal* that signed-link prediction exploits — structural balance — is exactly a *holonomy*: signs are a $\mathbb{Z}_2$ connection and balance is its trivial holonomy. (iii) HSiKAN is naturally read as a learned *section* of a fiber bundle over the hypergraph; a *rotor* (a Spin element) on each arc is a *connection* that parallel-transport those learned fibers; *spike-walks* (the tropical/event reading) time the transport; and the *spiral* collects the transported fibers at the walk ends, computing a learned, continuous holonomy that *strictly generalises the balance parity*. This predicts a concrete, falsifiable way the construction could lift AUROC — distinct from the rotor-as-readout experiment already settled in the negative — and simultaneously opens an event-based (neuromorphic) regime and a structural debuggability we have already built. We state the mathematics, the architecture, the testable claims, and an honest scope (theorem vs. conjecture vs. engineering).

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1 Motivation: why a gauge picture

HSiKAN (Highway Signed KAN) has, on serial-chain control and on Bitcoin-OTC signed-link prediction, *tied* a parameter-matched MLP / B-spline baseline. Two readings are possible: “the structural prior is inert” or “the prior was tested where the structure carries no exploitable signal.” This report argues the second, and makes it precise: the predictive content of a signed (hyper)graph is *gauge-theoretic* — it lives in the *holonomy* of a connection, not in node features — so a method only benefits when (a) the topology has non-trivial holonomy to read (cycles, frustration, branching) and (b) the architecture actually computes a holonomy rather than a pooled feature. We show HSiKAN supplies a learned fiber, a rotor supplies a connection, spikes supply the timing, and the three compose into a holonomy readout that is the natural continuous lift of the balance invariant.

2 Preliminaries

2.1 Signed hypergraph and incidence

A *signed weighted hypergraph* is $H = (V, E, w, \sigma)$, $|V| = n$, $|E| = m$, with arc weights $w_{ev} \in \mathbb{R}$ and signs $\sigma_{ev} \in \{\pm 1\}$ for $v \in e$. Its star-expansion promotes each hyperedge to a hub and yields the *signed incidence* $B \in \mathbb{R}^{n \times m}$, $B_{ve} = \sigma_{ev} w_{ev}$ if $v \in e$ else 0. B is the discrete boundary $\partial : C_1 \rightarrow C_0$ of the chain complex; $B^\top$ the coboundary δ .

2.2 Fiber bundles, connections, holonomy (discrete)

Definition 1 (Discrete G -bundle and connection). Let G be a group (the *structure group*) acting on a fiber F . A *discrete G -connection* on the graph assigns to each oriented arc $u \rightarrow v$ an element $g_{uv} \in G$ with $g_{vu} = g_{uv}^{-1}$. A *section* assigns to each vertex a fiber element $h_v \in F$. *Parallel transport* of h_u along $u \rightarrow v$ is $g_{uv} \cdot h_u$.

Definition 2 (Holonomy). For a walk $W = (v_0, \dots, v_k)$ the *transport* is $\text{Hol}(W) = g_{v_{k-1}v_k} \cdots g_{v_0v_1} \in G$. For a closed walk (cycle) $\text{Hol}(W)$ is the *holonomy*; it is invariant under *gauge transformations* $g_{uv} \mapsto s_v g_{uv} s_u^{-1}$ for any vertex relabelling $s : V \rightarrow G$ (up to conjugacy).

2.3 Rotors (Clifford / Spin)

In the Clifford algebra $\text{Cl}(\mathbb{R}^d)$ a *rotor* $R = e^{\frac{1}{2}\theta \hat{B}}$ ($\hat{B}$ a unit bivector) is an even, unit-norm element; $R \in \text{Spin}(d)$ acts on a vector x by the sandwich $x \mapsto RxR^{-1}$, a rotation by θ in the plane $\hat{B}$. Rotor composition (the geometric product) is associative, unit-bearing ($R = 1$), and *non-commutative* for $d \geq 3$ — the property that lets an ordered product encode order.

3 The semiring convolution (the computational substrate)

Definition 3 (Aggregation algebra). A commutative monoid $(S, \oplus, \mathbf{0})$; for signed-weighted aggregation a semiring $(S, \oplus, \otimes, \mathbf{0}, \mathbf{1})$ with $\otimes$ distributing over $\oplus$ and $\mathbf{0}$ annihilating.

Definition 4 (Convolution). For a field $x \in S^V$, $(B \circledast x)_e = \bigoplus_{v \in e} \sigma_{ev} w_{ev} \otimes x_v$ — the semiring matrix–vector product.

Proposition 1 (Well-definedness). $(B \circledast x)_e$ is order-independent over the unordered members of e iff $\oplus$ is commutative and associative; defined on empty/masked members iff $\oplus$ has identity; composable across nesting iff $\otimes$ distributes over $\oplus$. Hence a tag is admissible iff it names a commutative monoid (semiring for signed-weighted).

Remark 1 (Ordered walks vs. unordered members). A hyperedge’s members are unordered, so the cross-member fold needs a *commutative* $\oplus$. A *walk* is ordered, so transport along it is a chain of $\otimes$; there $\otimes$ may be non-commutative (a rotor) and this is desirable — it encodes walk order. The two never conflict: commutativity is required only of $\oplus$.

Catalog of regimes. additive $(\mathbb{R}, +, \times)$ — the graph Laplacian / linear message passing; parity $(\{\pm 1\}, \times)$ — Cartwright–Harary balance; tropical $(\mathbb{R} \cup \{\infty\}, \min, +)$ — event/spike timing; fuzzy $([0, 1], \perp, \top)$ — residuated, single-layer unless distributive; **spinor** (Spin, $\oplus$, geometric product) — the connection/transport regime developed below.

4 Signed graphs are $\mathbb{Z}_2$ gauge fields; balance is holonomy

Theorem 1 (Balance as trivial holonomy). Take $G = \mathbb{Z}_2 = \{\pm 1\}$ with the sign σ_{uv} as the connection. A signed graph is balanced (Cartwright–Harary: every cycle has an even number of negative edges, i.e. $\prod_{e \in C} \sigma_e = +1$) iff its $\mathbb{Z}_2$ holonomy is trivial, $\text{Hol}(C) = +1$ for every cycle C ; equivalently iff the connection is gauge-equivalent to the all-+1 connection ($\exists s : V \rightarrow \{\pm 1\}$ with $s_u \sigma_{uv} s_v = +1$).

Sketch. $\text{Hol}(C) = \prod_{e \in C} \sigma_e$ is the product of signs around C ; gauge transformations cancel in pairs around any closed walk, so Hol is gauge-invariant, and $\text{Hol} \equiv +1$ on a spanning set of cycles is exactly solvability of $s_u \sigma_{uv} s_v = +1$ on a spanning tree extended by the cycle constraints. $\square$

Remark 2 (Where the AUROC signal lives). Real signed graphs are *frustrated*: $\text{Hol}(C) \neq +1$ on some cycles. Signed-link prediction is precisely the problem of inferring an edge’s sign from the holonomy of surrounding cycles; the *frustration* (deviation of Hol from +1) is the residual signal beyond a globally balanced fit. Any predictor is, implicitly, estimating holonomy.

5 The Spin lift: from $\mathbb{Z}_2$ to a learned continuous holonomy

Replace $G = \mathbb{Z}_2$ by $G = \text{Spin}(d)$ (or $U(1)/SU(2)$): each arc carries a *rotor* $R_{uv} \in \text{Spin}$, a *learned* connection, and $\pi : \text{Spin} \rightarrow \mathbb{Z}_2$ (e.g. the sign of a fixed scalar component, or restriction to $\{\pm 1\} \subset \text{Spin}$) recovers the discrete sign.

Proposition 2 (The Spin holonomy refines balance). Under $\pi : \text{Spin} \rightarrow \mathbb{Z}_2$, $\pi(\text{Hol}_{\text{Spin}}(C)) = \text{Hol}_{\mathbb{Z}_2}(C) = \prod_{e \in C} \sigma_e$. Thus the rotor holonomy $\text{Hol}_{\text{Spin}}(C) \in \text{Spin}$ is a continuous invariant whose $\mathbb{Z}_2$ projection is exactly the balance parity; it carries strictly more information (a rotation angle / bivector plane), and it is gauge-covariant.

Remark 3. Hol_{Spin} is the discrete *curvature* (Wilson loop) of the learned connection: balanced cycles give $\text{Hol}_{\text{Spin}} \approx 1$, frustrated cycles give a non-trivial rotation whose magnitude grades the frustration. This is the object we propose to feed the predictor — a learned, continuous generalisation of the exact balance signal.

6 HSiKAN is the fiber; the rotor is the connection

Definition 5 (HSiKAN section). HSiKAN’s per-vertex output $h_v \in \mathbb{R}^d$ (the `node_activations` before pooling) is a learned section of the bundle: a Catmull-Rom-activated, highway-gated, signed-conv aggregation. Stacking layers is a learned *covariant* aggregation (message passing along B).

The construction has two stages, deliberately separated (this is what distinguishes it from the settled rotor-as-readout experiment):

1. **Fiber (HSiKAN):** learn rich per-vertex representations h_v — the heavy representational work.
2. **Connection (rotor):** transport those fibers along walks via R_{uv} ; *collect the transported fibers at the walk ends*; the readout consumes $\text{Hol}_{\text{Spin}}(W) \cdot h$, not raw signs.

7 Walk-spikes: the tropical timing of the transport

Read arc weights as *delays*: the tropical semiring $(\min, +)$ computes the *earliest spike arrival* along signed walks; a walk W emits a spike at $\tau_W = \sum_{e \in W} w_e$ carrying its parity (and, in the Spin lift, its rotor $\text{Hol}_{\text{Spin}}(W)$). A depth- k stack is a leaky-integrate-and-fire pass with structural delays.

Remark 4 (Why timing matters here). The spike *gates which transport reaches the head first*: the tropical min selects the *shortest* signed walk, and short walks accumulate less sign noise, so the earliest-spike readout is a principled “trust the shortest balanced path” bias. Idempotence $a \oplus a = a$ makes the representation sparse and event-driven — the neuromorphic / embedded regime, and the natural form of the *fast reflex* in the rate-asymmetric controller.

8 The synthesis: the spiral collects fibers at spike-walk ends

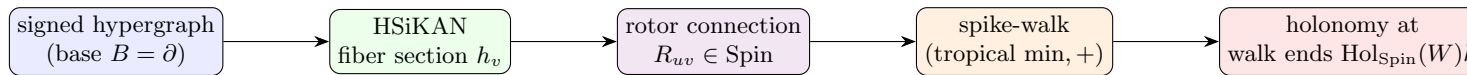


Figure 1: HSiKAN learns the fiber; the rotor connection transports it; spikes time the transport (shortest signed walk first); the spiral collects the parallel-transported fibers at the walk ends; a fixed real-bilinear head scores. The new signal is the *transport/holonomy*, with the readout held at the algebra already shown best.

The edge (u, v) is scored from the holonomy-transported fibers along $u \rightarrow v$ walks,

$$\hat{y}_{uv} = f_{\text{bilinear}} \left(\bigoplus_{W: u \rightarrow v} \alpha_W \text{Hol}_{\text{Spin}}(W) \cdot h \right),$$

where α_W is the (tropical-gated) walk weight and $\oplus$ the commutative cross-walk fold (Rem. 1). The code carried to the head is three-part: **sign** ($\mathbb{Z}_2$ parity), **timing** (tropical τ_W), **ordered phase** (Spin holonomy).

9 Testable claims and honest scope

- T1** (*theorem*, Thm. 1, Prop. 2) Balance = $\mathbb{Z}_2$ holonomy; the rotor holonomy projects onto it and refines it. *Established*.
- C1** (*conjecture, the AUROC bet*) A holonomy-transport readout — rotor connection on the HSiKAN fiber, collected at spike-walk ends, **readout fixed at real-bilinear** — lifts OTC AUROC over the flat (additive, transport-free) readout, because it estimates the frustration the flat code discards. *Untested; distinct from the settled rotor-as-readout experiment (which it does not re-run)*.
- C2** (*near-certain, efficiency*) The tropical/spike reading is sparse and event-driven; event-rate $\ll$ dense activations. *Measurable*.
- C3** (*conjecture, structure-complexity*) HSiKAN beats a params-matched MLP in proportion to topological complexity (branching/cycles/repetition); the discriminating test is quadruped / k -agent, not a serial chain. *Untested*.
- E1** (*engineering, done*) Because the fiber is named by structure, failures localise to a component (`hsikan_diagnose`); HSiKAN is debuggable where an MLP is not. *Built + tested*.

What is settled in the negative. The rotor/quaternion/geodesic algebra *as the readout metric on raw cycle features* did not beat real bilinear (the readout algebra is not the bottleneck). C1 therefore moves the rotor into the *connection/transport*, leaving the readout fixed — a different mechanism, not a repeat.

10 Differential and spectral structure (per regime)

Additive: $L = BB^\top$, Hodge $C_1 = \text{im } \delta \oplus \text{im } \partial \oplus \mathcal{H}$; potential-based reward shaping = the exact 1-forms. Parity: $\mathbb{Z}_2$ cohomology; balance = coboundary (Thm. 1). Tropical: max-plus eigenvalues / Bellman operator. Spinor: a non-abelian connection, holonomy = curvature (Wilson loop); no inner-product Hodge theory — the relevant invariant is the holonomy group. The algebra chosen is the (co)homology computed.

11 Relation to prior lines

Signed-triad attention / Berge cycles (the readout study — the settled negative); the Cayley/rotor primitive (here re-rolled from metric to connection); the FuzzySignature (the per-cycle σ -vote = the $\mathbb{Z}_2$ holonomy read-out the structural critic already computes); the structural critic (the per-motif value = a holonomy aggregate); the reachability-audit line (A^* = reachability over the base, planner demos as exact transport); the rate-asymmetric controller (spikes = the event-based fast reflex).

12 Implementation path

1. The **aggregate** semiring extension (declared algebra per relation; the rotor is the spinor- $\otimes$ on ordered walks). Grammar is CORE — escalation-gated; default **sum**.
2. A rotor-connection layer (per-arc R_{uv} , $O(d^2)$ params) over the existing HSiKAN fiber; the holonomy readout collects $\text{Hol}_{\text{Spin}}(W)h$ at walk ends (walks already enumerated by `hymeko_graph`).
3. Tests: C1 holonomy-vs-flat on OTC (readout fixed); C2 spike sparsity; C3 quadruped HSiKAN-vs-MLP; E1 `hsikan_diagnose` (done). Parity guard: $\pi(\text{Hol}_{\text{Spin}}) = \text{BalanceScorer}$.

13 Open questions and risks

Does Hol_{Spin} overfit frustration noise (regularise toward $\text{Hol} \approx 1$)? Is the gauge-covariance of the readout enforced or merely hoped (a non-gauge-covariant head can wash out the holonomy — prefer gauge-invariant features, e.g. $\|\log \text{Hol}_{\text{Spin}}\|$ or class traces)? Does the non-distributive fuzzy regime break nesting (yes — mark single-layer). Is the AUROC plateau representational or label-noise-limited (the leakage-audit line)?

One-paragraph summary

Signed-hypergraph computation is a semiring convolution; the predictive signal is holonomy; balance is its $\mathbb{Z}_2$ special case; HSiKAN is the fiber, a rotor the connection, spikes the timing, and the spiral collects the parallel-transported fibers at the spike-walk ends to form a learned continuous holonomy that generalises the balance parity. That is a falsifiable reason it could lift AUROC (with the readout held fixed), an event-based regime for free, and — already in hand — a structurally debuggable controller.